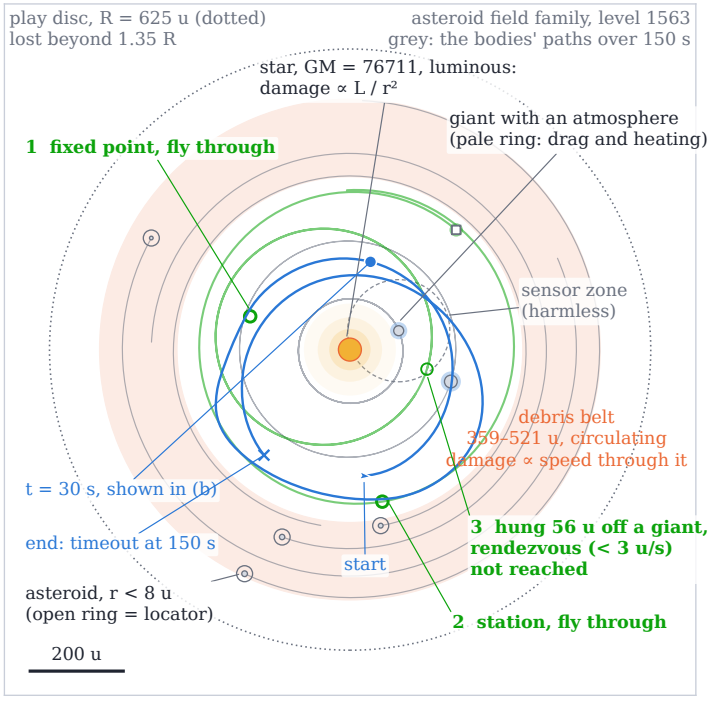
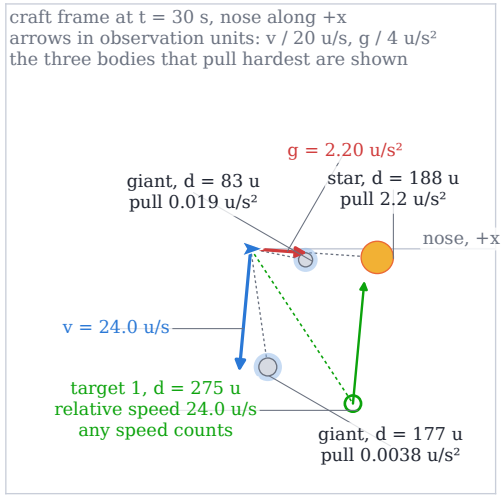
(a) the level with a sampled rollout

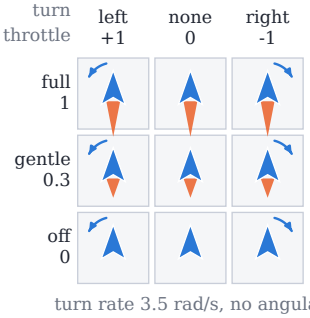


(b) what the policy observes



one body token (24 numbers) per body, up to 32:
relative position, direction, distance, relative velocity, its pull ÷ the engine, GM, radius, atmosphere, luminosity, clearance, is-a-target, kind
one zone token (15) per zone, up to 12:
where the band is, its radii, signed distance to it, kind, strength, spin
self block (109):
v, g, the current target's offset, distance and relative velocity; fuel, Δv left, Δv left ÷ Δv the tour still needs, engine, health, time left, targets pending, order-free, damage rate, exposure, distance ÷ R, the objective weights, 4 field-readout slots (Sec. 3); 13 per target slot (×6)
tokens are pooled by attention with the self block as the query, so nothing depends on how many bodies there are or their order

(c) the nine actions



turn rate 3.5 rad/s, no angular inertia; thrust = throttle × engine acceleration; an action is held for one tick (67 ms)

(d) goal kinds

fly-through



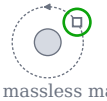
pass within 12 u, at any speed

rendezvous



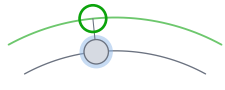
pass within 12 u, and |v craft − v target| < 3 u/s

station



a massless marker on its own orbit; fly through or rendezvous

attached to a body



rides a planet or moon at a fixed offset; fly through or rendezvous

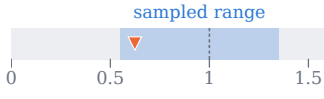
a task is a tour of 1 to 6 such targets, each a ring of radius 12 u, visited in sequence or, when the task says so, in any order

(e) loadout

engine a ∈ [0.6, 3] u/s², log-uniform

tank f ∈ [3, 60] s of full burn

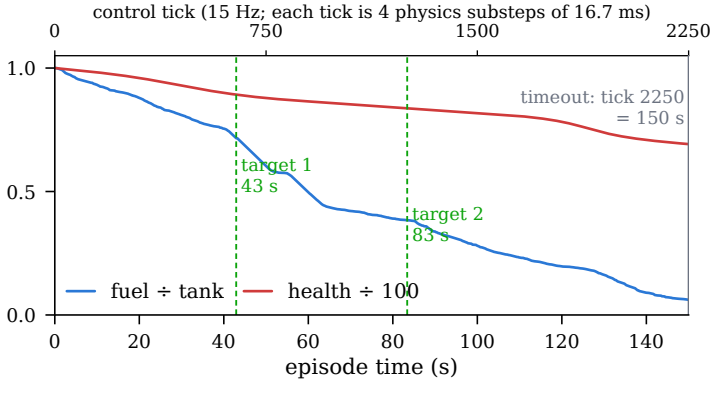
Δv = a · f, with the tank sized so that Δv ÷ a direct-flight estimate is uniform in [0.55, 1.35], then clipped to the tank limits



Δv ÷ direct-flight estimate

this task (▾): a = 0.94 u/s², f = 60 s, Δv = 57 u/s against ≈ 91 u/s for a direct flight, a ratio of 0.62

(f) the episode clock



units

u game length unit (a level is 1000–3000 u across)
s game second; physics at 60 Hz, the pilot at 15 Hz
G = 1, so a body's mass is its GM, in u³/s²; a massless body (station, tracer) is carried by gravity, exerts none
speeds in u/s, accelerations in u/s²; health 0–100, damage in /s

(g) endings and units

how an episode ends (checked every substep)

arrived	every target reached; the only success
crashed	inside a solid body's radius
destroyed	health ≤ 0: star radiation, atmospheric heating, debris, radiation belts
lost	farther than 1.35 R from the centre
timeout	tick 2250 = 150 s, tour unfinished
tank empty	not an ending: the craft coasts on